



PARALLEL COMPUTER ARCHITECTURE

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PRE-REQUISITES : Final year undergraduate as well as post-graduate students in the disciplines of CSE, IT, CS, ECE, EEE, etc. Can benefit from this course. A basic knowledge in Computer Organisation is sufficient to take this subject.

INTENDED AUDIENCE : Final year UG as well as PG

INDUSTRY SUPPORT : Intel, Micron, AMD, IBM, NVIDIA, HP, Samsung, etc.

COURSE OUTLINE :

With the emergence of application domains requiring large data processing as well as faster compute performance, parallel computing has become a critical component. This involves the use of multi-core processors as well as tightly integrated clusters of PCs and workstations. The most exciting development is the convergence of traditionally disparate approaches of shared memory, message passing, and data driven computing onto the same platform.

As the parallel applications communicate with each other using shared data; it is important to correctly manage the shared data within the memory hierarchy to avoid inconsistency. Here, we need to understand the mechanisms used for data access, communication and coordination of work and its correct implementation.

This course will introduce the field of parallel architectures and discuss in-depth shared memory management for parallel architectures. We will also learn about the interconnection topologies and routing methods which are important for connection and communication of the multiple cores executing the parallel applications. The course will conclude with an overview of various parallel architectures and on-going research directions.

ABOUT INSTRUCTOR :

Prof. Hemangee K. Kapoor is a Professor in the Department of Computer Science Engineering at the Indian Institute of Technology Guwahati, Assam. She has over 18 years of teaching experience at both undergraduate and postgraduate levels. She received her B.Eng. degree in Computer Engineering from the College of Engineering, Pune, the M.Tech. degree in CSE from the IIT Bombay, and the Ph.D. degree in CSE from London South Bank University, U.K. Prof. Kapoor has been teaching subjects related to computer organisation, digital design, advanced computer architecture, parallel computer architecture and hardware lab. She has also given expert lectures in FDPs and TEQIP events. Her research spans in the broad area of computer architecture and specially in multicore caches, emerging non-volatile memory technologies, network-on-chip and design of accelerators for neural networks. Till date she has graduated 6 PhD and 39 MTech students. She is a Senior Member of the IEEE and ACM. She has published her work in reputed journals and conferences. She is an Associate Editor for the Journal of Systems Architecture and IEEE Design and Test. Prof. Kapoor has served in the ACM India council: as a member (2014-22) and as Vice President (2020-22). She serves as a member on the ACM international council for Diversity, Equity and Inclusion (ACM-DEI) (2021-23). More details can be found on website: URL: <https://iitg.ac.in/hemangee/>

COURSE PLAN :

Week 1 : Introduction to Parallel Architectures

Week 2 : Parallel Programming models and Architectures

Week 3 : Memory Hierarchy-Cache and Virtual memory

Week 4 : Overview of Cache coherence, Coherence Protocols- Snooping, Directory based protocols, VI protocol

Week 5 : MSI, MESI, Dragon protocol and Correctness of coherence protocols- Types of cache misses, update vs invalidate protocol

Week 6 : Snoop based multiprocessor design, Single and multi-level cache with atomic bus

Week 7 : Snoop based multiprocessor design, Single and multi-level cache with split transaction bus

Week 8 : Scalable shared memory systems: Directory coherence protocols- Memory based, cache based, correctness

Week 9 : Case study: Origin- Architecture, protocol, correctness; Sequent NUMA Q- Architecture, protocol, correctness

Week 10 : Memory consistency models- Sequential, Relaxed consistency models

Week 11 : Synchronization- LL-SC, point to point, barrier synchronization

Week 12 : Interconnects- Introduction, Topologies, routing, flow control